

PCB: Content Upside & PTFE Adoption

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What's new

We view Nvidia's Rubin platform as a significant catalyst for PCB content value expansion, with PTFE adoption potentially driving the next leg of the upgrade cycle into Rubin Ultra. We expect PCB content value to increase from ~US\$440 per GPU for GB300 to ~US\$750 for VR200 and potentially ~US\$1,000–1,500 for VR300 (Oberon), driven by higher-end CCL materials, increasing layer counts, and the potential PTFE adoption.

Comments

VR200 PCB content value to reach US\$750 per GPU, up from US\$440 for GB300: We expect VR200 to adopt M8+M6 CCL for the compute tray, while the switch tray to migrate to M8.5 CCL (Low-DK2/HVLP4). The new 44L midplane is also expected to adopt M8.5 CCL. These material and design upgrades should raise PCB content value to ~US\$750 per GPU, from ~US\$440 for GB300.

VR300's PCB content to reach \$1k with further upside on PTFE: We believe Nvidia is still finalizing the PCB specifications for VR300, with final decisions likely to be made by end-26. The key area of debate is the new switch tray design for the Oberon Rack.

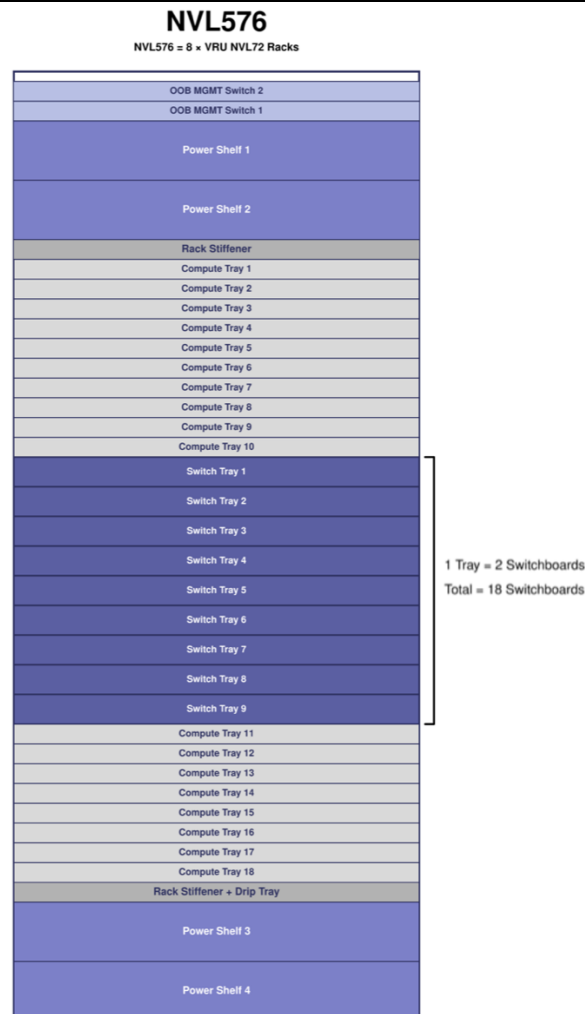
- We expect the VR300 NVL72 (non-expandable version) compute tray to upgrade to M8.5 (Low-DK2 + HVLP4) + M6 hybrid, while the switch tray and midplane are likely to migrate to M9 (Q-glass). This should further lift VR300 PCB content value to ~US\$1,000 per GPU.
- For the VR300 NVL576 Oberon Rack, we see an upgrade in both switch board design (increasing to 18 switch boards) and CCL materials. Multiple solutions are under consideration, mostly based on 16+16L designs and centered on hybrid PTFE. Potential configurations include PTFE + M9 (Low-DK2/Q-glass) or M10 (Q-glass), while the use of glass-fiber cloth versus glass-free prepreg for the PTFE layers remains under consideration. For glass-free prepreg, two approaches are currently being evaluated: 1) chopped glass fibers dispersed in resin; and 2) Hydrocarbon film/ABF film to replace PP. We believe the PTFE switch board is more likely to be adopted given its significantly lower layer count versus the orthogonal midplane, which translates into meaningfully higher yields. If adopted, PTFE could raise PCB content value to more than US\$1,500 per GPU for the VR300 Oberon Rack.

Stock preference: 1) PCB suppliers: we favor **ZDT (4958 TT, Buy)**, given its newly established position in the VR200 compute tray supply chain; we also expect Chinese leading player to sustain high share in compute and switch. **2) For PTFE CCL,** we believe Chinese leading PTFE CCL player is currently the most advanced at all front and will be the primary supplier; The **EMC (2383 TT, Buy) +Taiflex (8039 TT)** combination could be a potential second-source candidate, with PTFE CCL samples delivered to **Unimicron (3037 TT)** for further PCB development.

Risks

1) AI demand slowing down; 2) Geo-political risks; 3) Competition.

Figure 1: VR300 Oberon Rack Design



Sources: Nvidia Company Data, GF Securities (Hong Kong) Brokerage

Figure 2: Nvidia PCB & CCL SPEC

Platform	Baseboard	CCL Level	CCL Material	CCL \$ per GPU	PCB \$ per GPU
GB300 Bianca	Compute	M8+M4	DQN	145	440
	Switch	M8	892K2		
VR200	Compute	M8+M6	HVLP4+DK1	250	750
	Midplane	M8.5	896K2		
	Switch	M8.5	896K2		
VR300	Compute	M8.5+M6	HVLP4+Low DK2	350	1000
	Midplane	M9	896K3		
	Switch	M9	896K3		
VR300 Oberon	Compute	M8.5+M6	HVLP4+Low DK2	~ 500	~ 1500
	Midplane	M9	896K3		
	Switch	PTFE + M9/M10	5300 + ?		

Sources: Company Data, GF Securities (Hong Kong) Brokerage

Rating definitions Benchmark: Hang Seng Index (Hong Kong)

Company ratings	Buy	Stock expected to outperform benchmark by more than 10%
	Hold	Expected stock relative performance ranges between -10% and 10%
	Underperform	Stock expected to underperform benchmark by more than 10%

Sector ratings	Positive	Sector expected to outperform benchmark by more than 10%
	Neutral	Expected sector relative performance ranges between -10% and 10%
	Cautious	Sector expected to underperform benchmark by more than 10%

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